Ordering Fractions and Mixed Numbers

Ordering with benchmarks, common denominators, and mixed numbers

Directions:

- Choose your tool for each problem: a benchmark like $\frac{1}{2}$ or 1, a common denominator, or rewriting a mixed number.
- When a blank asks for a symbol, write <, >, or =.

1. Write the correct symbol: $\frac{5}{8}$ ________ $\frac{1}{2}$

2. Write $\frac{7}{4}$ as a mixed number.

Answer: ________________

3. Order from least to greatest by rewriting everything in sixths: $\frac{2}{3}$, $\frac{1}{2}$, $\frac{5}{6}$

________ $<$ ________ $<$ ________

4. Use the benchmark $\frac{1}{2}$ to compare $\frac{3}{8}$ and $\frac{5}{9}$.

$\frac{3}{8}$ ________ $\frac{1}{2}$ $\frac{5}{9}$ ________ $\frac{1}{2}$ so $\frac{3}{8}$ ________ $\frac{5}{9}$

5. On a number line, point P sits at $2\frac{1}{3}$. Write the location of P as an improper fraction in thirds.

Answer: ________________

6. Order from least to greatest (twenty-fourths make a good common denominator): $\frac{7}{8}$, $\frac{3}{4}$, $\frac{5}{6}$

________ < ________ < ________

7. Order from least to greatest. Rewrite each as eighths first: $1\frac{3}{4}$, $\frac{15}{8}$, $1\frac{1}{2}$

________ < ________ < ________

8. Three hiking trails: Pine is $\frac{5}{12}$ mi, Oak is $\frac{4}{6}$ mi, and Elm is $\frac{1}{2}$ mi. Order the trail lengths from least to greatest. *Hint: compare each to* $\frac{1}{2}$.

________ < ________ < ________

9. Ben says: "$2\frac{1}{4} < \frac{9}{4}$, because 2 is less than 9."
(a) Write $2\frac{1}{4}$ in fourths: __________
(b) Write the correct symbol: $2\frac{1}{4}$ __________ $\frac{9}{4}$
(c) Explain what Ben overlooked.

10. Order from least to greatest (thirty-sixths work for all three): $\frac{5}{6}$, $\frac{7}{9}$, $\frac{8}{12}$
__________ < __________ < __________

11. Order from least to greatest using the benchmarks 1 and $1\frac{1}{2}$: $\frac{13}{8}$, $1\frac{1}{4}$, $\frac{3}{2}$
__________ < __________ < __________

12. Challenge. Three ribbons: ribbon A is $1\frac{5}{6}$ m, ribbon B is $\frac{11}{6}$ m, and ribbon C is $1\frac{2}{3}$ m.

(a) Compare A and B: $1\frac{5}{6}$ __________ $\frac{11}{6}$

(b) Compare C and B: $1\frac{2}{3}$ __________ $\frac{11}{6}$

(c) Which ribbon is the shortest? Say why two of the ribbons are a tie.